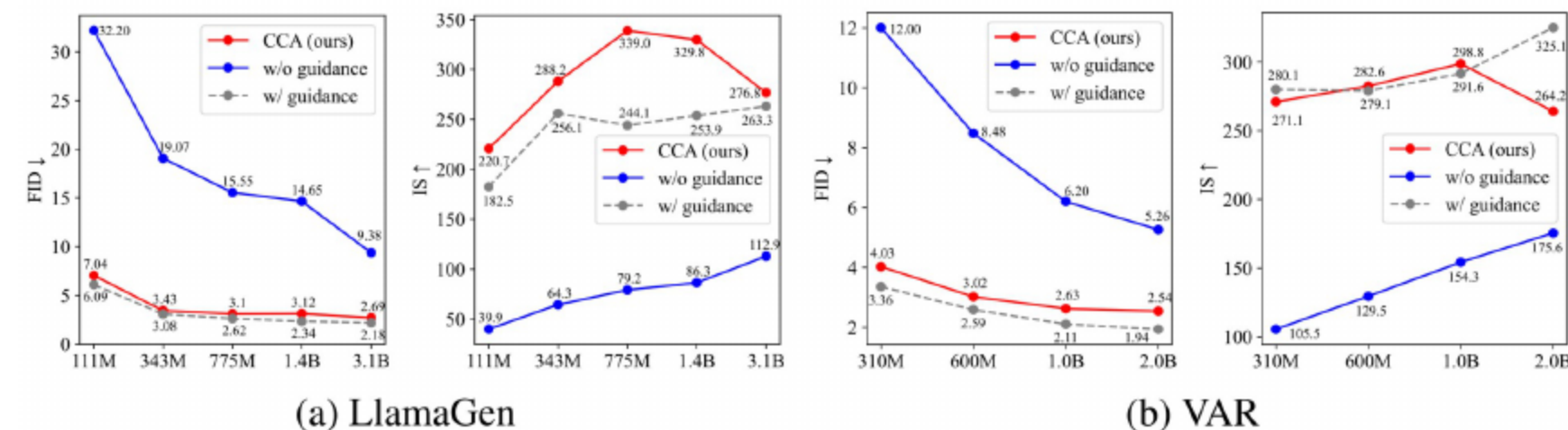


Toward Guidance-Free AR Visual Generation via Condition Contrastive Alignment

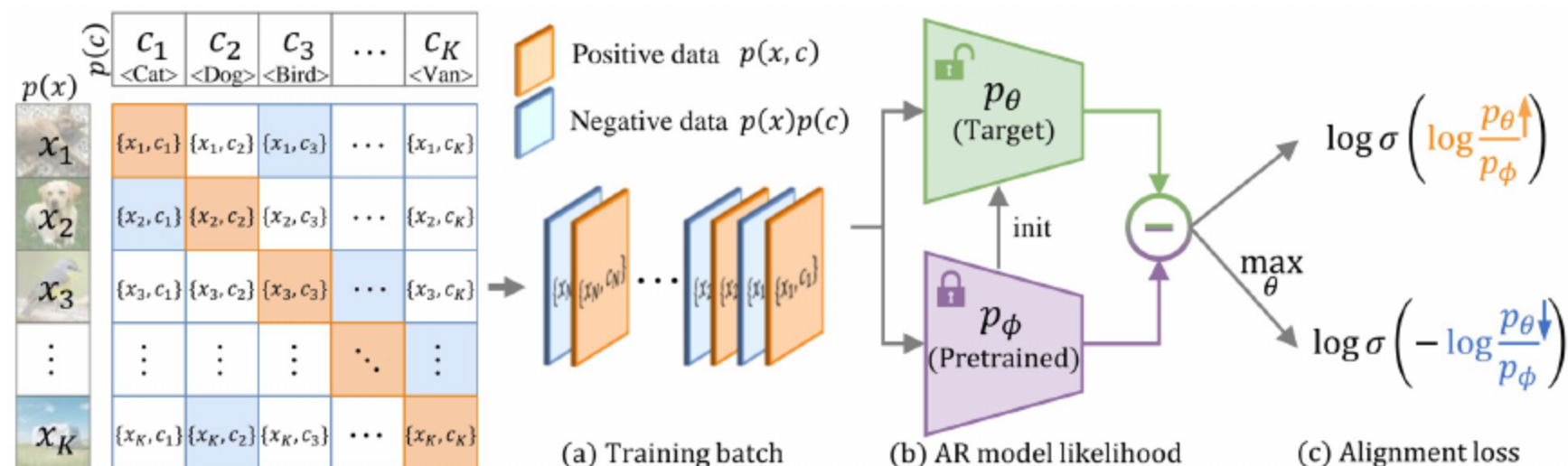
Huayu Chen, Hang Su, Peize Sun, Jun Zhu

01 Unifying AR Visual Generation with LLMs



- **AR vision as next-token prediction** unifies with LLM-style modeling across modalities
- Visual AR still relies on Classifier-Free Guidance, while text LMs use guidance-free decoding
- **Guidance doubles sampling cost** and requires masked-condition training, complicating pipelines
- Key question: **Can fine-tuning learn guidance effects** and remove guided sampling? [Esser et al., 2021; Lee et al., 2022; Team, 2024] [Xie et al., 2024; Ho & Salimans, 2022]

02 Our Approach and Contributions



- **Condition Contrastive Alignment:** guidance-free AR via contrastive fine-tuning on pretraining data
- Construct positives as matched pairs and negatives by mismatching conditions within a batch
- **Efficient fine-tuning:** about one epoch, single pass sampling, no extra datasets
- Unifies LM alignment and visual guidance by directly learning the conditional residual effect

03 AR Modeling and Visual Tokenization

- Autoregressive learning predicts the next token in a sequence given prior tokens
- Images are discretized with vector-quantized tokenizers; encoder and decoder are trained then frozen
- This allows AR models to process image tokens in the same manner as text tokens

- Sampling mixes conditional and unconditional logits using a tunable guidance scale
- Improves fidelity and diversity** but requires two forward passes per token
- Unconditional behavior is learned by randomly masking prompts during training

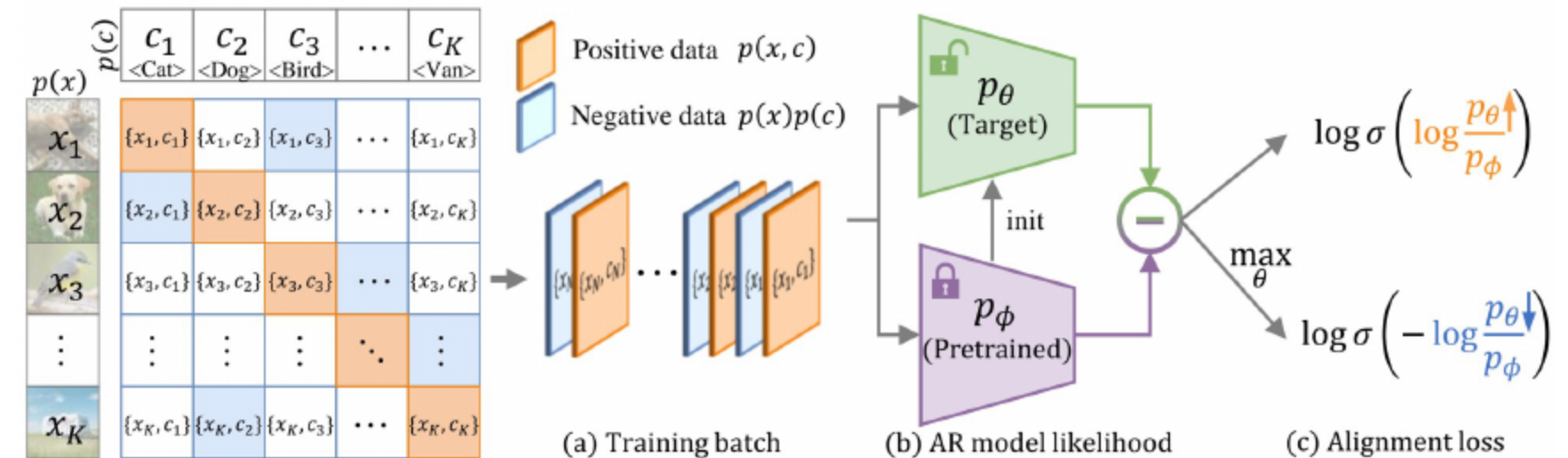
Method	Classifier Guidance	Classifier-Free Guidance	Condition Contrastive Alignment
Modeling of $\log \frac{p(c x)}{p(c)}$	$\log p_\theta(c x) - \log p_\theta(c)$	$\log p_\theta(x c) - \log p_\theta(x)$	$\beta [\log p_\theta^{\text{sample}}(x c) - \log p_\theta(x c)]$
Training loss	$\max_{\theta} \mathbb{E}_{p(x,c)} \log p_\theta(c x)$	$\max_{\theta} \mathbb{E}_{p(x,c)} \log p_\theta(x)$	$\min_{\theta} \mathcal{L}_\theta^{\text{CCA}}$ in Eq. 11
Sampling policy	$\log p_\theta(x c) + s \log p_\theta(c x)$	$(1+s) \log p_\theta(x c) - s \log p_\theta(x)$	$\log p_\theta^{\text{sample}}(x c)$
Extra training cost	$\sim 9\%$ of learning p_θ	$\sim 10\%$ of learning p_θ	$\sim 1\%$ of pretraining p_θ
Sampling cost	$\times \sim 1.3$	$\times 2$	$\times 1$
Applicable area	Diffusion	Diffusion & Autoregressive	Autoregressive

[Ho & Salimans, 2022; Podell et al., 2023; Chang et al., 2023]
[Sun et al., 2024]

- Guidance targets a distribution shift captured by the conditional residual between conditional and unconditional probabilities
- Core idea: directly learn this residual using a contrastive objective inspired by noise contrastive estimation
- Removes the need for an extra unconditional model and specialized decoding

[Gutmann & Hyvärinen, 2012]

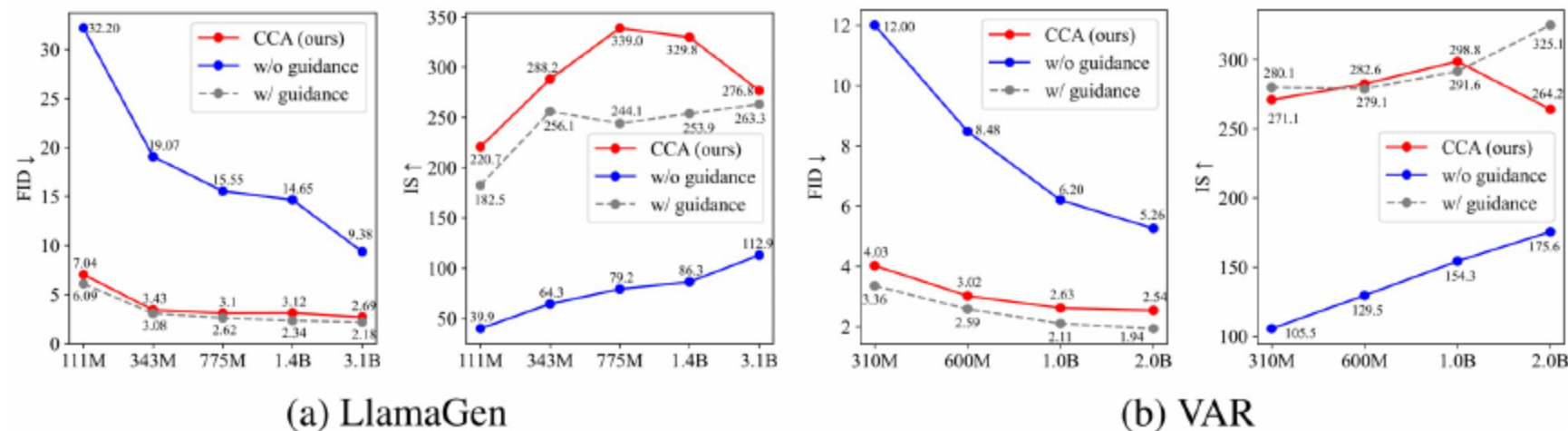
- Represent the residual as the change in log likelihood between a learnable target model and a frozen pretrained model
- Train a single autoregressive sampler with standard decoding while keeping the reference model frozen
- Initialize the target model from the pretrained checkpoint; no unconditional branch is required



- Build batches with matched pairs and negatives by shuffling conditions across the batch
- Use two training weights that mirror guidance scale and control the penalty on negatives
- One epoch of fine-tuning is typically sufficient** for strong improvements

[Rafailov et al., 2023; Chen et al., 2024a]

08 Teaser: CCA Delivers CFG-like Quality Gains



- Brief fine-tuning yields major quality gains without guided sampling
- Matches much of the guidance benefit for autoregressive models using standard decoding
- Enables controllable trade-offs through training hyperparameters

09 Main Quantitative Results on ImageNet 256

	Model	w/o Guidance				w/ Guidance	
		FID↓	IS↑	Precision↑	Recall↑	FID↓	IS↑
Diffusion	ADM (Dhariwal & Nichol, 2021)	7.49	127.5	0.72	0.63	3.94	215.8
	LDM-4 (Rombach et al., 2022)	10.56	103.5	0.71	0.62	3.60	247.7
	U-ViT-H/2 (Bao et al., 2023)	—	—	—	—	2.29	263.9
	DiT-XL/2 (Peebles & Xie, 2023)	9.62	121.5	0.67	0.67	2.27	278.2
	MDTv2-XL/2 (Gao et al., 2023)	5.06	155.6	0.72	0.66	1.58	314.7
Mask	MaskGIT (Chang et al., 2022)	6.18	182.1	<u>0.80</u>	0.51	—	—
	MAGVIT-v2 (Yu et al., 2023)	3.65	200.5	—	—	1.78	319.4
	MAGE (Li et al., 2023)	6.93	195.8	—	—	—	—
Autoregressive	VQGAN (Esser et al., 2021)	15.78	74.3	—	—	5.20	280.3
	ViT-VQGAN (Yu et al., 2021)	4.17	175.1	—	—	3.04	227.4
	RQ-Transformer (Lee et al., 2022)	7.55	134.0	—	—	3.80	323.7
	LlamaGen-3B (Sun et al., 2024)	9.38	112.9	0.69	0.67	2.18	263.3
	+CCA (Ours)	<u>2.69</u>	276.8	<u>0.80</u>	0.59	—	—
	VAR-d30 (Tian et al., 2024)	5.25	175.6	0.75	0.62	1.92	323.1
	+CCA (Ours)	2.54	<u>264.2</u>	0.83	0.56	—	—

- Record guidance-free FID for autoregressive models on ImageNet at 256 resolution
- Strong Inception Score improvements across LlamaGen and VAR families
- Closes most of the gap to classifier-free guidance while retaining single pass sampling

10 Modeling the Guidance Term: Unified Comparison

Method	Classifier Guidance	Classifier-Free Guidance	Condition Contrastive Alignment
Modeling of $\log \frac{p(\mathbf{x} \mathbf{c})}{p(\mathbf{x})}$	$\log p_\theta(\mathbf{c} \mathbf{x}) - \log p(\mathbf{c})$	$\log p_\phi(\mathbf{x} \mathbf{c}) - \log p_\theta(\mathbf{x})$	$\beta[\log p_\theta^{\text{sample}}(\mathbf{x} \mathbf{c}) - \log p_\phi(\mathbf{x} \mathbf{c})]$
Training loss	$\max_\theta \mathbb{E}_{p(\mathbf{x}, \mathbf{c})} \log p_\theta(\mathbf{c} \mathbf{x})$	$\max_\theta \mathbb{E}_{p(\mathbf{x})} \log p_\theta(\mathbf{x})$	$\min_\theta \mathcal{L}_\theta^{\text{CCA}}$ in Eq. 11
Sampling policy	$\log p_\phi(\mathbf{x} \mathbf{c}) + s \log p_\theta(\mathbf{c} \mathbf{x})$	$(1 + s) \log p_\phi(\mathbf{x} \mathbf{c}) - s \log p_\theta(\mathbf{x})$	$\log p_\theta^{\text{sample}}(\mathbf{x} \mathbf{c})$
Extra training cost	$\sim 9\%$ of learning p_ϕ	$\sim 10\%$ of learning p_ϕ	$\sim 1\%$ of pretraining p_ϕ
Sampling cost	$\times \sim 1.3$	$\times 2$	$\times 1$
Applicable area	Diffusion	Diffusion & Autoregressive	Autoregressive

- Classifier guidance, classifier-free guidance, and CCA all target the same conditional residual
- CCA trains the final sampling model directly, matching standard language model decoding
- Lower compute and simpler pipelines compared with auxiliary guidance models

11 Experimental Questions and Setup

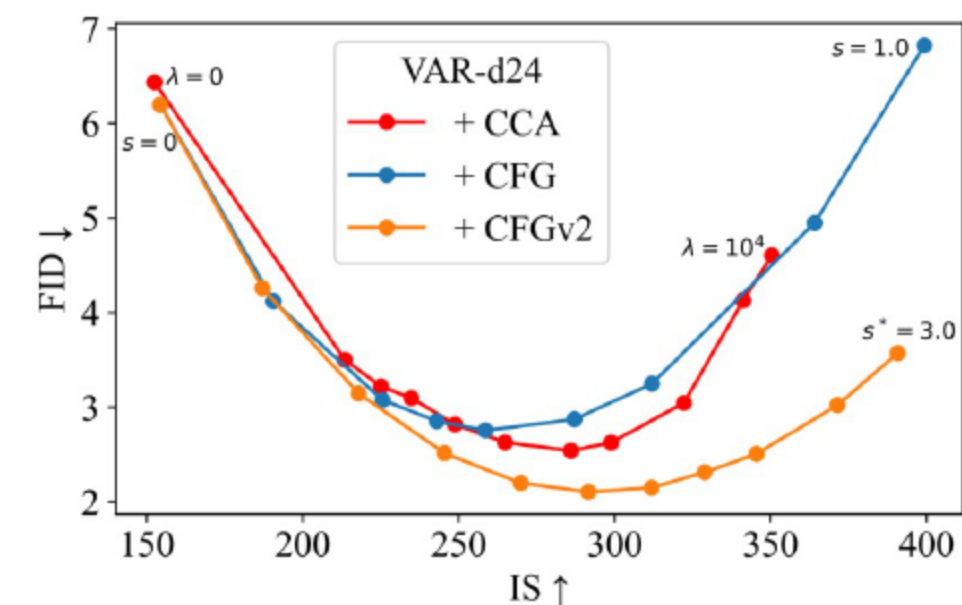
- Questions: effectiveness, fidelity versus diversity, comparisons to language model alignment, and synergy with guidance
- Models: LlamaGen and VAR with distinct tokenizers and architectures on ImageNet
- Training: one epoch fine-tuning, fix the main residual weight, tune the negative weight per model, guidance off at inference

12 Headline Improvements and Qualitative Examples



- Large FID and Inception Score gains after one epoch of fine-tuning
- Better class alignment and fidelity without any guidance at sampling time
- Comparable to guided quality while keeping single pass sampling

13 Diversity-Fidelity Trade-off: CCA vs. CFG



- Sweeping the negative loss weight reproduces FID and Inception Score trade-offs like guidance scale
- While the residual weight matters, the negative weight is the primary practical knob
- The curves closely resemble standard guidance; scheduled guidance variants show similar behavior

[Chang et al., 2023]

14 CCA vs. LLM Alignment Methods

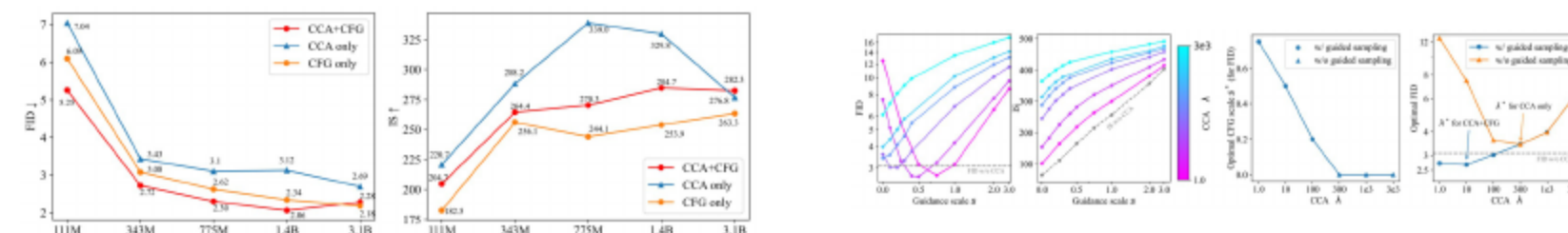
Model	FID↓	IS	sFID↓	Precision	Recall
LlamaGen-L	19.00	64.7	8.78	0.61	0.67
+DPO	61.69	30.8	44.98	0.36	0.40
+Unlearn	12.22	111.6	7.99	0.66	0.64
+CCA	3.43	288.2	7.44	0.81	0.52

Model	FID↓	IS	sFID↓	Precision	Recall
VAR-d24	6.20	154.3	8.50	0.74	0.62
+DPO	7.53	232.6	19.10	0.85	0.34
+Unlearn	5.55	165.9	8.41	0.75	0.61
+CCA	2.63	298.8	7.63	0.84	0.55

- Direct preference optimization can underperform or collapse on autoregressive visual likelihoods
- Including a frozen reference model is crucial to balance positive and negative likelihoods
- Naive preference transfer is not well matched to token-level visual likelihood training

[Welleck et al., 2019; Rafailov et al., 2023; Chen et al., 2024a]
[Pal et al., 2024]

15 CCA + CFG: Integration and Hyperparameters



- CCA plus guidance often outperforms guidance alone with tuned negative weight
- Include unconditional positives with masked prompts to align the unconditional branch
- Optimal negative weight differs for guidance-free versus combined training; smaller values help the combo

- Language model alignment evolved from reinforcement learning with feedback to direct objectives for single stage training
- Visual generation: adversarial and diffusion models excel on pixels; autoregression with tokenization enables LM-style modeling
- Prior visual alignment often needs extra data; CCA reuses pretraining data and targets replacing guidance at sampling

[Schulman et al., 2017; Ouyang et al., 2022; Rafailov et al., 2023]
[Chen et al., 2024a; Ho et al., 2020]



THANKS!

- [CCA trains guidance effects into autoregressive models](#), enabling single pass sampling with near guided quality
- Offers a controllable fidelity and diversity trade-off through training weights
- Code: <https://github.com/thu-ml/CCA>